Behavioural and Mechanistic Measurements Diverge During Small-Model Training

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Abstract

Behavioural evaluations and mechanistic measurements provide different views of learned model capabilities, but it is unclear whether these measurements stabilize on the same timeline during training. We study this question in a controlled induction-learning setting by training small decoder-only transformers from scratch across four random seeds and jointly tracking behavioural accuracy and a causal activation-patching measurement over 8,000 training steps.

Behavioural induction performance reaches ceiling within approximately 300–500 steps across all four seeds and remains there for the remainder of training. The activation-patching measurement follows a different trajectory, exhibiting substantial seed-level variability beyond step 1,000 before reaching a sustained near-ceiling plateau later in training. Under a sustained-ceiling criterion evaluated across thresholds $\theta \in [0.80, 1.00]$, behavioural stabilization occurs at $T_b = 300$, whereas mechanistic stabilization occurs at $T_m = 2{,}000$–$2{,}500$ for 20 of 21 tested thresholds, corresponding to a measured separation of 1,700–2,200 training steps.

A matched negative control does not reproduce the sustained late-training patching plateau, and head-level localization reveals heterogeneous mechanistic organization across random seeds. An architecture robustness probe further identifies a configuration with perfect behavioural performance but a substantially weaker patching effect. A complementary real-text experiment confirms substantial language-model learning but does not reproduce the clean depth-dependent diagnostic observed in prior induction-head work at the tested scale and training budget.

These results establish that behavioural performance and a causal mechanistic measurement of the same learned capability can stabilize on different timelines in a controlled small-model setting. They do not establish that the underlying mechanism itself is necessarily unstable whenever the probe changes. Instead, the findings identify a measurement distinction: behavioural stability alone does not imply that a particular mechanistic measurement has stabilized or will recover the learned capability consistently.

1 Introduction

Model capabilities can be studied from at least two complementary perspectives. Behavioural evaluation measures what a model does under specified inputs and conditions, whereas mechanistic methods attempt to characterize the internal computations causally responsible for that behaviour. Although these measurements are often interpreted together, they need not evolve identically during training.

This distinction matters for model evaluation. A behavioural score may stabilize while the internal computation supporting that behaviour continues to reorganize, while multiple circuit implementations may support similar observable behaviour, or while a mechanistic probe may vary in its sensitivity across training. In each case, stable behavioural performance and stable mechanistic measurement would convey different information about the learned capability.

We investigate this measurement problem using induction learning as a controlled experimental setting. Induction heads provide a particularly useful test case because the target behaviour is precisely defined, can be learned by small transformers, and admits causal investigation through activation patching. Rather than asking whether induction heads exist or how they implement in-context learning, we ask a longitudinal measurement question:

> **When behavioural performance and a causal mechanistic probe track the same learned capability across training, do they tell the same story about when that capability has stabilized?**

We train a four-layer decoder-only transformer from scratch for 8,000 steps across four random seeds and measure both behavioural induction accuracy and an activation-patching effect throughout training. Behavioural performance reaches ceiling early, whereas the patching measurement remains substantially more variable before reaching a sustained near-ceiling plateau later.

Using a sustained-ceiling criterion, we measure a separation of 1,700–2,200 training steps between behavioural and mechanistic stabilization across 20 of 21 tested threshold values. We complement this result with a matched negative control, attention-head localization, an architecture robustness probe, and a real-text extension.

The contribution of this work is therefore primarily methodological. We provide a controlled empirical example in which behavioural and causal mechanistic measurements of the same learned capability provide different pictures of training dynamics. The result motivates treating behavioural and mechanistic measurements as complementary evidence rather than assuming that stability in one necessarily implies stability in the other.

2 Related Work

2.1 Induction Heads

Olsson et al. (2022) identified induction heads as a causal mechanism supporting in-context copying behaviour in transformer language models. Their work characterized the emergence of induction heads during training, connected their formation to phase changes in model loss, and used mechanistic interventions to establish their causal contribution to in-context learning.

2.2 Causal Interventions and Activation Patching

The activation-patching methodology used here follows a broader line of work on causal interventions in neural network internals. Meng et al. (2022) use causal tracing to identify internal states that contribute to model predictions by restoring selected activations from an uncorrupted run within a corrupted computation. Our experiments apply the same general intervention logic at the level of transformer layers and attention heads, measuring how much of the clean-versus-corrupted logit gap is recovered after patching. Related causal-mediation approaches have also been used to attribute model behaviour to specific internal components (Vig et al., 2020).

We use activation patching as a measurement tool rather than proposing a new patching technique. Our contribution is to study how this causal measurement evolves relative to behavioural performance over the course of training.

The present study builds on that experimental setting but asks a different question. Rather than characterizing the induction circuit itself, we use induction as a tractable capability for which behavioural and causal mechanistic measurements can be repeatedly observed throughout training.

This allows us to examine whether the two measurements stabilize together across checkpoints and random seeds.

Our focus is therefore not on establishing the existence or generality of induction heads, but on the relationship between behavioural and mechanistic measurements during capability acquisition.

3 Experimental Setup

3.1 Synthetic Induction Task

Each training sequence is drawn from one of two conditions. With probability 0.7, a randomly sampled context chunk is followed by an exact repetition of that chunk, creating a learnable induction structure. With probability 0.3, the sequence consists entirely of unstructured random tokens.

The vocabulary contains 100 tokens and sequences have length 64. This synthetic construction makes induction-predictable positions precisely identifiable while retaining unstructured sequences that prevent the training distribution from consisting entirely of repeated patterns.

3.2 Model and Training

The main model is a pre-LayerNorm decoder-only transformer with four layers, four attention heads per layer, model dimension $d_{\text{model}} = 128$, and feed-forward dimension $d_{\text{ff}} = 512$.

Models are trained from scratch for 8,000 optimization steps across random seeds 0, 1, 2, and 3. Checkpoints are recorded every 50 steps through step 1,500 and every 500 steps thereafter.

3.3 Behavioural Measurement

Behavioural performance is measured as next-token prediction accuracy at induction-predictable positions in sequences containing the repeated-chunk structure. With a 100-token vocabulary, chance accuracy is 0.01.

3.4 Mechanistic Measurement

At each checkpoint, we construct matched clean and corrupted sequence pairs and measure the fraction of the clean-versus-corrupted logit gap recovered when attention outputs from the clean execution are patched into the corrupted execution.

The primary analysis measures layer-level patching effects, while a subsequent localization analysis measures individual attention heads. The probed location corresponds to a genuine single-token induction position.

3.5 Stability Criterion

An initial analysis considered low cross-seed variance for three consecutive checkpoints as evidence of stability. This criterion was rejected because it can classify uniformly poor, floor-level performance as stable. In the present data, this produced the invalid result $T_b = 0$ despite behavioural induction performance emerging only after training began.

We therefore define stability using a sustained-ceiling criterion. For a measurement M and threshold θ, the stabilization checkpoint is

$$T(M, \theta) = \min \{t : M_s(t_i) \geq \theta \ \ \forall s, \ \ i \in \{0, 1, 2\}\},$$

where t, t_1, and t_2 are three consecutive recorded checkpoints beginning at t, and the condition must hold for every random seed s.

We evaluate this criterion over

$$\theta \in \{0.80, 0.81, \ldots, 1.00\}.$$

This threshold sweep reduces dependence on any single chosen ceiling value.

4 Results

4.1 Behavioural and Mechanistic Measurements Stabilize on Different Timelines

Behavioural induction accuracy reaches 1.000 across all four seeds within approximately 300–500 training steps and remains at ceiling thereafter. The activation-patching measurement follows a markedly different trajectory, with substantial seed-level fluctuations continuing beyond step 1,000 before a sustained near-ceiling plateau emerges later in training.

Under the sustained-ceiling criterion, behavioural stabilization occurs at

$$T_b = 300,$$

while mechanistic stabilization occurs at

$$T_m \in \{2000, 2500\}$$

for thresholds $\theta = 0.80$ through $\theta = 0.99$. The resulting measured separation is therefore

$$T_m - T_b = 1700\text{–}2200$$

training steps.

The ordering

$$T_b < T_m$$

holds for 20 of the 21 thresholds examined. At $\theta = 1.00$, T_m does not resolve because the patching measurement oscillates within approximately ± 0.0005 of the exact numerical ceiling rather than remaining at precisely 1.000 for three consecutive checkpoints. We therefore interpret this edge case as sensitivity to floating-point precision rather than evidence of a qualitatively different training trajectory.

These results establish a threshold-robust measured separation between behavioural and activation-patching stabilization in this experimental setting.

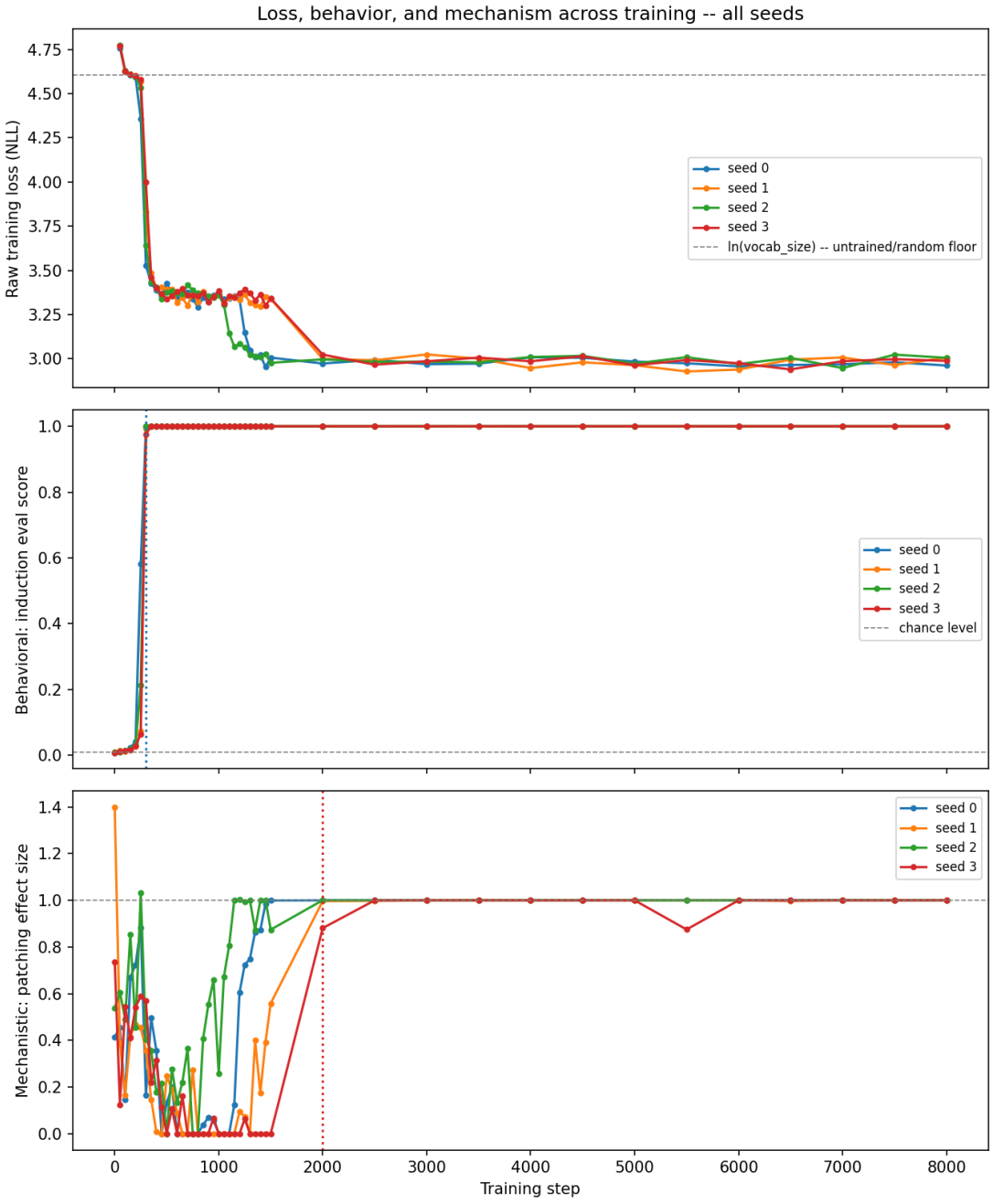


Figure 1: Training loss, behavioural induction performance, and activation-patching effect across four random seeds. Behavioural performance reaches ceiling early in training, whereas the mechanistic measurement remains substantially more variable before reaching a sustained near-ceiling plateau later.

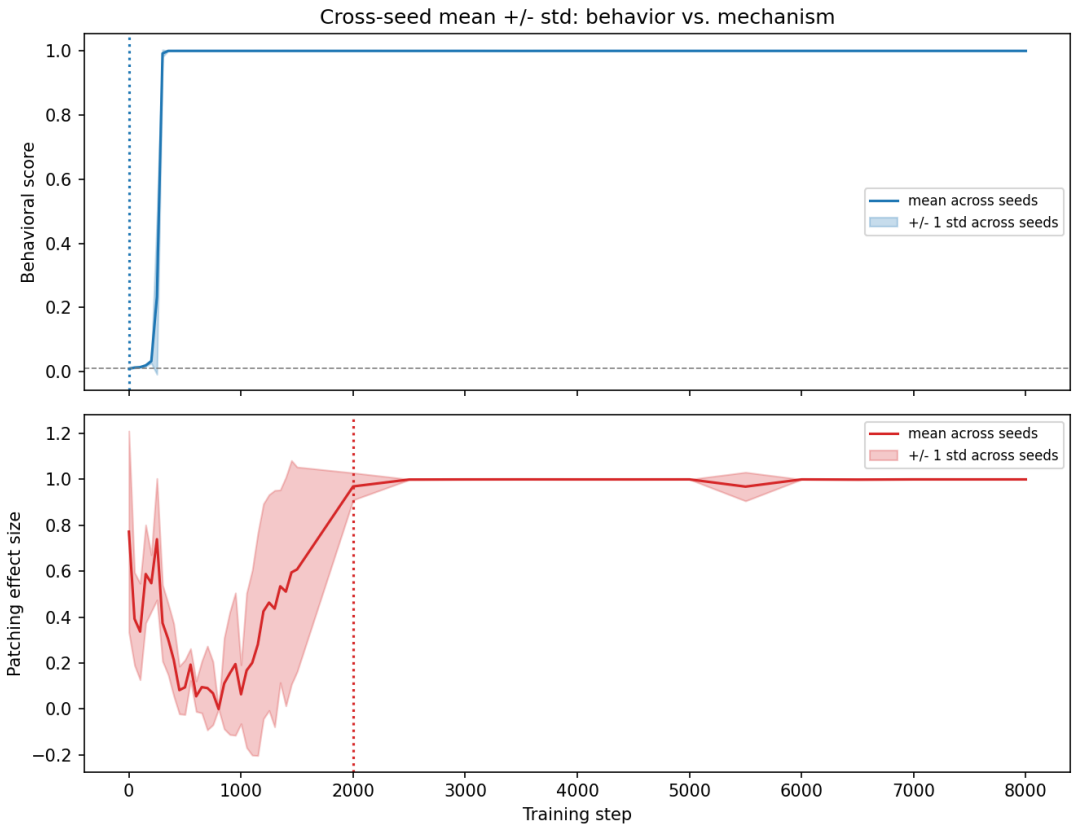


Figure 2: Cross-seed mean and standard deviation for behavioural induction performance and activation-patching effect across training. Behavioural performance rapidly converges to ceiling, while the mechanistic measurement exhibits substantially greater cross-seed variability during the intermediate training regime.

4.2 Negative Control

To test whether the patching pipeline itself could generate the sustained pattern observed in the main experiment, we train otherwise identical models with the induction structure removed entirely.

These models do not acquire the behavioural induction task. The patching measurement nevertheless exhibits transient early effects, reaching values as high as 0.732, before decaying to approximately zero by step 1,500 and remaining there.

The negative control therefore provides evidence that the sustained late-training patching plateau in the main condition is not explained solely by baseline behaviour of the measurement pipeline. It does not eliminate all possible measurement artifacts.

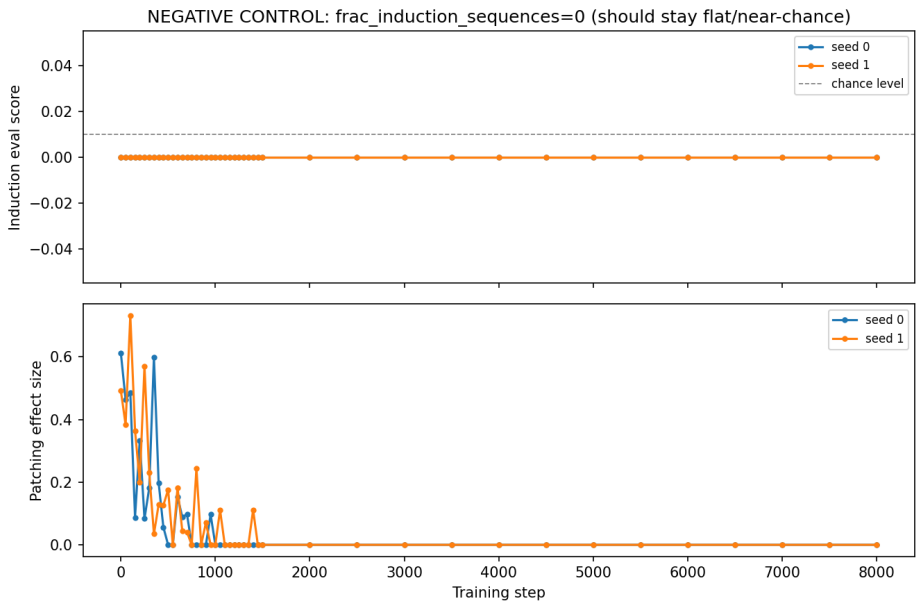


Figure 3: Negative-control trajectories for models trained without induction-structured sequences. Behavioural induction performance remains at zero, while transient activation-patching effects decay rather than forming the sustained late-training plateau observed in the main condition.

4.3 Head-Level Localization

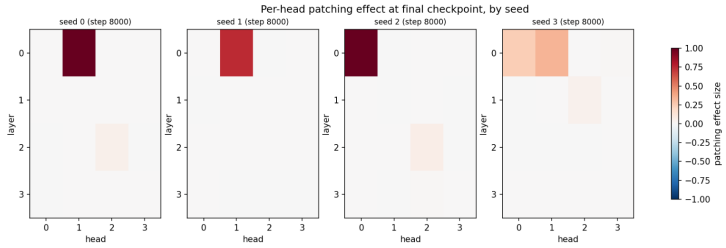


Figure 4: Per-head activation-patching effects at the final checkpoint for each random seed.

Three of four seeds concentrate most of the final patching effect in a single layer-0 attention head. The remaining seed exhibits a more distributed pattern spanning multiple heads and layers.

This heterogeneity indicates that similar behavioural performance does not require identical localization under the present mechanistic measurement. It also motivates caution when interpreting a single head-level decomposition as a canonical implementation of the learned behaviour.

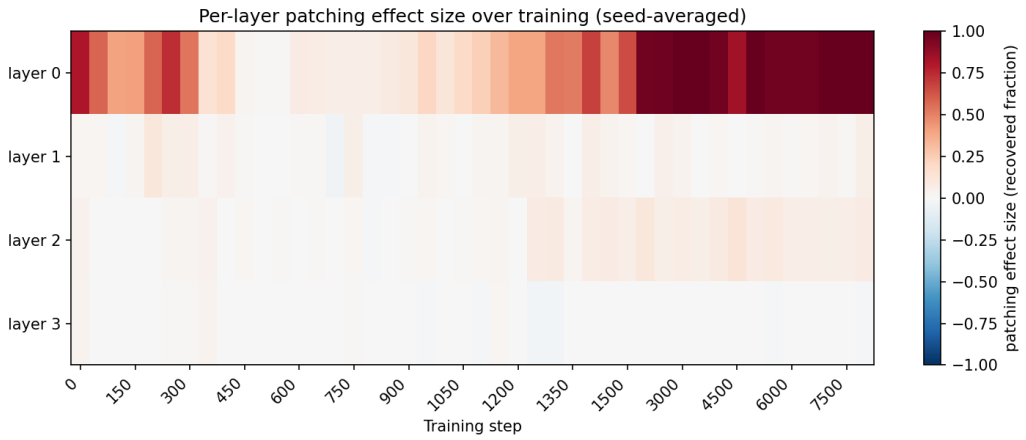


Figure 5: Seed-averaged activation-patching effect by layer over training. The strongest sustained late-training effect localizes primarily to layer 0, while later layers contribute substantially less under this measurement.

4.4 Architecture Robustness Probe

Table 1: Final behavioural and activation-patching measurements across tested architectures.

d_{model}	Layers	Behavioural score	Patching effect
64	2	1.000	0.088
128	4	1.000	1.000
256	4	1.000	1.000

All three tested configurations reach perfect behavioural accuracy by step 4,000, but their final activation-patching measurements differ substantially. Most notably, the $d_{\text{model}} = 64$, two-layer configuration achieves behavioural accuracy of 1.000 while exhibiting a patching effect of only 0.088.

Because both width and depth change in this configuration, the experiment does not isolate either variable as the cause of the discrepancy. The result instead identifies an architecture-dependent behavioural/probe mismatch that warrants controlled follow-up experiments.

5 Real-Text Extension

The primary experiment uses a synthetic task to make the target capability and its behavioural measurement precisely identifiable. We additionally test related diagnostics in a natural-language setting using attention-only character-level models trained on the public-domain text of Jane Austen's *Pride and Prejudice*.

5.1 Depth-Dependent Diagnostic

At the tested model scale and 3,000-step training budget, the one-, two-, and three-layer models do not reproduce the clean depth-dependent pattern reported in the original induction-head work. We retain this null result because it constrains the extent to which the synthetic findings can presently be connected to natural-language training.

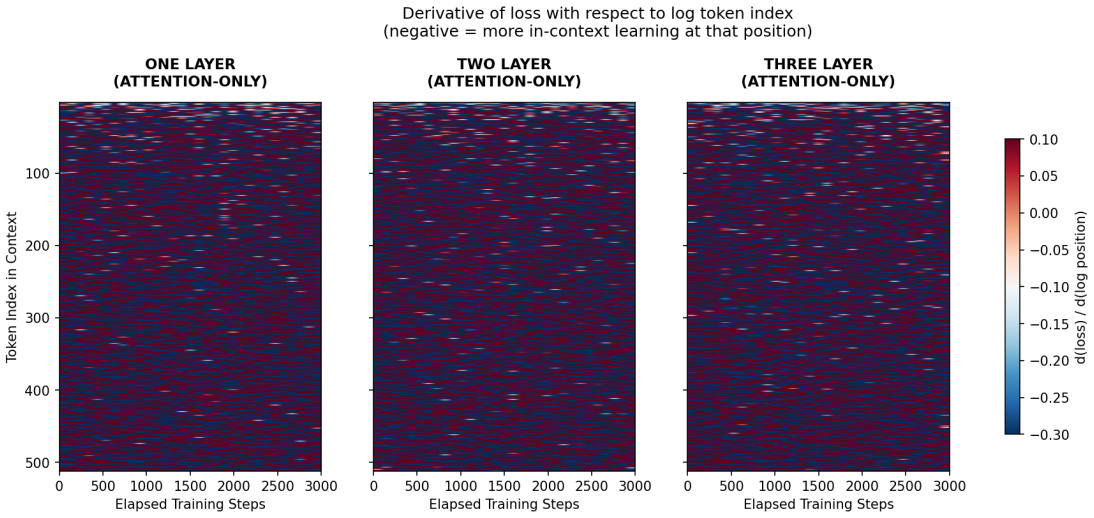


Figure 6: Derivative-of-loss diagnostic for one-, two-, and three-layer attention-only character models trained on *Pride and Prejudice*.

5.2 Prediction Improvement on Held-Out Text

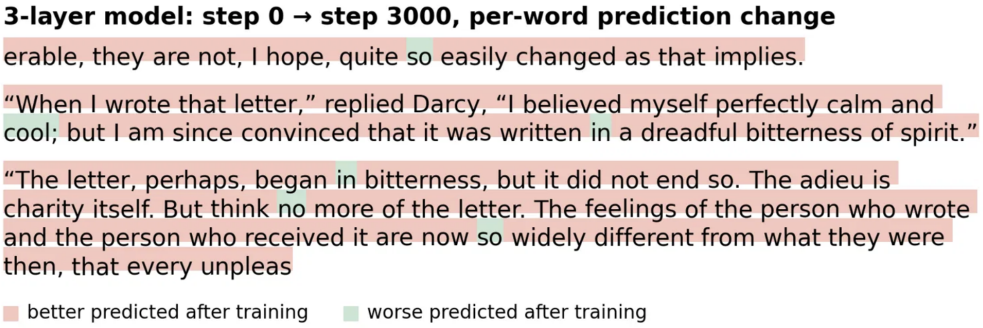


Figure 7: Character-level prediction changes between initialization and step 3,000 for a held-out passage from *Pride and Prejudice*.

For the three-layer model, 93.7% of positions in a held-out passage improve meaningfully under a loss-improvement threshold of 0.15 nats, while 4.9% worsen. Mean loss decreases from 4.626 at initialization to 2.443 after training.

This result demonstrates substantial real-text learning. It does not establish that an induction circuit causally produced the observed improvements; doing so would require mechanistic intervention in the real-text models.

6 Robustness and Measurement Diagnostics

6.1 Threshold Sensitivity

The sustained-ceiling analysis produces $T_b < T_m$ across every threshold from $\theta = 0.80$ through $\theta = 0.99$, corresponding to a measured separation of 1,700–2,200 training steps.

A corrected seed-resampling bootstrap has not yet been performed. We therefore treat the threshold sweep as evidence that the measured separation is not an artifact of a single ceiling choice, rather than as a resampling-based confidence statement.

6.2 Patching Noise Floor

The negative control demonstrates that substantial isolated patching values can occur transiently even in the absence of induction learning. Individual mid-training patching measurements should therefore not be interpreted independently.

The principal mechanistic evidence is instead the emergence of a sustained late-training plateau, considered jointly with behavioural performance, the negative control, and head-level localization.

7 Discussion

7.1 What the Results Establish

The central empirical result is a separation between the training dynamics reported by behavioural and mechanistic measurements. Behavioural induction performance reaches ceiling early and remains stable, whereas the activation-patching measurement continues to vary before reaching a sustained near-ceiling plateau substantially later.

This separation persists across a broad sweep of stability thresholds and is not reproduced by the matched negative control.

7.2 What the Results Do Not Establish

The observed patching variability should not be interpreted as direct evidence that the underlying mechanism itself remains unstable throughout the same period. Several explanations remain compatible with the data.

The model may reorganize how induction is implemented while preserving behaviour. Multiple circuit implementations may support equivalent behavioural performance. Alternatively, activation patching may differ in sensitivity across checkpoints or architectures.

The present experiment distinguishes the measurements; it does not fully distinguish among these mechanistic explanations.

7.3 Implications for Model Evaluation

Behavioural evaluation measures observable model performance. Mechanistic analysis provides a different form of evidence concerning the internal computations associated with that performance.

In this controlled setting, those measurements do not stabilize on the same timeline. Stable behavioural performance should therefore not, by itself, be interpreted as evidence that a particular mechanistic account has stabilized or that a chosen causal probe will recover the relevant computation consistently.

The implication is not that mechanistic measurements should replace behavioural evaluation. Rather, the experiment motivates treating behavioural, mechanistic, and robustness evidence as complementary measurements whose agreement should itself be empirically tested.

7.4 Limitations

The primary experiment uses a small transformer and a synthetic induction task. Induction is unusually tractable relative to complex capabilities in contemporary language models, and the magnitude or direction of the measured separation may not generalize beyond this setting.

The architecture probe also varies width and depth simultaneously, preventing causal attribution of the 64/2 result to either factor individually.

Finally, the real-text extension demonstrates language learning but does not yet include activation patching. It therefore cannot establish that the behavioural-mechanistic relationship observed in the synthetic task extends to natural-language induction circuits.

8 Future Work

The highest-priority follow-up is to resolve the architecture-dependent discrepancy by varying width and depth independently and performing component-level patching in the smallest configuration.

A second direction is to increase the training budget and dataset size for the real-text experiment and extend activation patching to those models.

A corrected seed-resampling bootstrap under the sustained-ceiling criterion would provide a resampling-based uncertainty estimate for the observed stabilization gap.

Finally, the same experimental logic can be applied to more complex learned behaviours to test whether behavioural and mechanistic measurements diverge outside the induction setting.

9 Conclusion

We studied whether behavioural and causal mechanistic measurements of the same learned capability stabilize together during training. In a controlled induction-learning experiment across four random seeds, behavioural performance reached ceiling early, while an activation-patching measurement remained substantially more variable and reached sustained near-ceiling values later. Under a sustained-ceiling criterion, the two measurements were separated by 1,700–2,200 training steps across 20 of 21 tested thresholds.

The result is intentionally narrow but methodologically important. Behavioural stability and mechanistic stability are distinct empirical properties, and agreement between them should not be assumed. In settings where conclusions depend on both what a model does and how that behaviour is implemented, measurement agreement is itself something to test.

References

[1] Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, et al. In-context learning and induction heads. *Transformer Circuits Thread*, 2022. `https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html`.

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[3] Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. Investigating gender bias in language models using causal mediation analysis. *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. `https://arxiv.org/abs/2004.12265`.